When Does Reranking Pay Off? A Cost-Aware Study of Chunking, Embedding, and Reranking Trade-offs in Retrieval-Augmented Generation

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***Abstract*—Retrieval-augmented generation (RAG) pipelines expose several configuration choices—chunk size, embedding model, and whether to apply cross-encoder reranking—each with different quality, latency, and cost implications. Reranking in particular is often assumed to matter more for multi-hop questions, where relevant evidence is scattered across passages and initial retrieval is more likely to be imperfect. We test this assumption directly with a controlled, reproducible study across a single-hop dataset (SQuAD v2) and a multi-hop dataset (HotpotQA), sweeping chunk size (256/512/1024 tokens) and reranking (on/off) with a fixed open-weight embedding model (BAAI/bge-small-en-v1.5) and generator (Llama-3.1-8B-Instant). Contrary to our hypothesis, we find no statistically significant effect of reranking on retrieval quality on either dataset (paired Wilcoxon, $p = 0.50$ and $p = 0.66$ respectively). Instead, chunk size is the dominant factor on the multi-hop dataset, nearly doubling retrieval Hit@k when increased from 256 to 1024 tokens, while reranking consistently adds 40–100%+ latency without a corresponding quality gain. We report this as an honest negative result and derive a simple practitioner takeaway: for multi-hop RAG under latency or cost constraints, increasing chunk size is a higher-leverage, lower-cost intervention than adding a reranking stage. All code, configurations, and seeded results are released for reproducibility.**

***Index Terms*—retrieval-augmented generation, reranking, chunking, cross-encoder, empirical study**

I. Introduction

Retrieval-augmented generation has become a default architecture for grounding large language models in external knowledge, but practitioners assembling a RAG pipeline face several under-examined design choices simultaneously: how large should each retrieved chunk be, which embedding model should encode them, and is it worth adding a reranking stage after initial retrieval? Reranking with a cross-encoder is widely recommended as a way to improve retrieval precision beyond what a bi-encoder can achieve alone, and the intuition is especially strong for multi-hop questions, where the evidence needed to answer a question is spread across multiple passages and a single bi-encoder similarity search is more likely to misrank the passages that matter.

This intuition, however, is rarely tested directly against its cost: reranking adds a full cross-encoder forward pass over every retrieved candidate, which is not free in latency or (for hosted rerankers) money. A practitioner deciding whether to add reranking to a production RAG pipeline needs to know not just *whether* it helps, but whether it helps *enough*, on *which kind* of question, *relative to its cost*—and whether cheaper interventions, like simply changing chunk size, might deliver more of the benefit for less.

We address this with a controlled factorial study: 3 chunk sizes $\times$ 2 reranking settings, evaluated on both a single-hop dataset (SQuAD v2) and a multi-hop dataset (HotpotQA, distractor setting), with 3 random seeds per configuration for variance estimation and a paired significance test isolating the reranking effect specifically. Our contribution is not a new retrieval or reranking method, but a rigorous, reproducible measurement of an assumption (reranking matters more for multi-hop) that is widely held but, to our knowledge, not directly quantified in this controlled a way at this scale in prior applied work. We report our finding honestly even though it runs against our stated hypothesis: reranking did not produce a statistically significant improvement on either dataset in this study, while chunk size did.

II. Related Work

RAG evaluation. Es et al. [1] introduce RAGAS, a reference-free evaluation framework for RAG pipelines that decomposes evaluation into retrieval quality (context precision/recall), faithfulness, and answer relevancy. Saad-Falcon et al. [12] propose ARES, an alternative automated RAG evaluation framework that instead fine-tunes lightweight LM judges on synthetic training data and corrects for judge error using prediction-powered inference against a small set of human annotations—trading RAGAS's zero-shot, reference-free simplicity for a more statistically calibrated but heavier evaluation pipeline requiring some human-labeled data upfront. Our study adopts a lighter-weight decomposition than either—separating retrieval-quality metrics (Hit@k) from answer-quality metrics (token-level F1)—relying on gold-passage matching rather than any LLM-judged scores for the retrieval side, trading both frameworks' flexibility and nuance for metrics that require no additional judge model, no synthetic training data, and no added cost or variance from judge-model calls.

Context utilization. Liu et al. [2] show that language model performance on long-context tasks degrades when relevant information sits in the middle of the context rather than at the beginning or end—the "lost in the middle" phenomenon.

This is directly relevant to the chunking and reranking choices studied here: reranking's implicit promise is that placing the most relevant chunk higher in the ranking should help the generator use it, and larger chunks risk diluting relevant content with more surrounding, less relevant text. Our finding that larger chunks *improved* rather than hurt multi-hop retrieval suggests that, at the token-length regime studied here (256–1024 tokens), a larger chunk's benefit outweighed any lost-in-the-middle penalty; this may not hold at substantially larger chunk sizes.

Chunk size and dataset type. Bhat et al. [11] systematically evaluate fixed-size chunking across multiple datasets and embedding models, finding that smaller chunks (64–128 tokens) are optimal for datasets with concise, fact-based answers, while larger chunks (512–1024 tokens) improve retrieval on datasets requiring broader contextual understanding. This closely parallels our own chunk-size finding: SQuAD v2 shows no benefit from larger chunks because retrieval is already near-ceiling at every size we tested, while HotpotQA shows a clear, substantial improvement from 256 to 1024 tokens—consistent with Bhat et al.'s claim that the right chunk size is dataset-dependent, not a universal constant. Our study extends this line of work by holding chunk size's interaction with a second factor, reranking, constant across the comparison, which their study does not examine.

Dense retrieval and embeddings. Karpukhin et al. [7] introduce Dense Passage Retrieval (DPR), establishing that a dual-encoder architecture trained contrastively can outperform sparse retrieval methods, underlying the bi-encoder retrieval stage used throughout our pipeline. Reimers and Gurevych [8] introduce Sentence-BERT, the architectural lineage the BAAI/bge-small-en-v1.5 model used in this study belongs to. Xiao et al. [3] introduce the BGE family of open-weight embedding models used here, evaluated on the C-MTEB/MTEB benchmarks. We use these models specifically because they are free to run locally, a hard constraint for this study's budget (see Limitations).

Cross-encoder reranking. Nogueira and Cho [9] demonstrate that a BERT-based cross-encoder substantially improves passage ranking accuracy on MS MARCO and TREC-CAR, establishing cross-encoder reranking as a standard second-stage retrieval-quality intervention—the specific mechanism our reranking condition tests via BAAI/bge-reranker-base. Our null result does not contradict their finding that cross-encoders can outperform bi-encoders in isolation on ranking benchmarks; rather, it suggests that in an end-to-end RAG pipeline where the generator only consumes the final top-3 chunks, the ranking-quality improvement a cross-encoder provides may not translate into a measurable downstream gain.

RAG system design. Lewis et al. [10] introduce the original Retrieval-Augmented Generation architecture, establishing the retrieve-then-generate paradigm this study's pipeline follows (without end-to-end fine-tuning). Gao et al. [6] survey the broader RAG landscape, observing that retrieval quality is frequently the primary bottleneck in RAG pipelines and that improvements to chunking and retrieval/reranking often matter more than swapping the generator model—a claim our design tests directly. Our result is a mixed confirmation: chunk size behaved as their synthesis would predict, but reranking did not show a measurable effect, suggesting the general claim does not hold uniformly across all retrieval-improving techniques.

III. Method

A. Pipeline

We implement a config-driven RAG pipeline with five stages: chunking, embedding, retrieval, optional reranking, and generation. All components are inference-only; no models are fine-tuned.

- **Chunking.** Source passages are split using a recursive character-based splitter with three chunk-size levels (256, 512, 1024 tokens) and a fixed 15% overlap.
- **Embedding.** Chunks and queries are embedded with BAAI/bge-small-en-v1.5, a compact open-weight bi-encoder, run locally on CPU. (An OpenAI text-embedding-3-small condition was originally planned as a paid-tier comparison point but was dropped from the final sweep—see Limitations.)
- **Retrieval.** A FAISS flat index performs cosine-similarity search, returning the top-5 chunks per query.
- **Reranking.** In the reranking condition, the top-5 retrieved chunks are re-scored by a BAAI/bge-reranker-base cross-encoder and the top-3 are kept; in the no-reranking condition, the top-3 of the original retrieval ranking are kept directly.
- **Generation.** Answers are generated by Llama-3.1-8B-Instant (served via Groq), temperature 0.

B. Experimental Design

We run a full factorial sweep over chunk size (3 levels) $\times$ reranking (2 levels), yielding 6 configurations per dataset. Each configuration is evaluated on 3 random seeds (42, 123, 456), each seed sampling 50 questions from a fixed pool of 150 questions per dataset, for up to 150 question-level observations per configuration. All other factors are held fixed across cells for fair comparison.

C. Datasets

- **SQuAD v2** (single-hop, factoid)—tests basic retrieval and extraction from a single gold passage per question.
- **HotpotQA**, distractor setting (multi-hop)—tests whether retrieval and reranking help when supporting evidence is spread across multiple passages mixed with distractors.

150 answerable questions were sampled from each dataset's validation split (seed=42) and held fixed across all configurations and seeds.

D. Metrics

- **Hit@k (retrieval).** 1 if the gold answer string appears verbatim in the concatenated text of the final retrieved chunks, 0 otherwise.

TABLE I
MEAN HIT@K, ANSWER F1, AND LATENCY BY CONFIGURATION

Dataset	Chunk	Rerank	Hit@k	F1	Latency (ms)	n
HotpotQA	256	No	0.380	0.331	1186	150
HotpotQA	256	Yes	0.400	0.355	2096	150
HotpotQA	512	No	0.544	0.389	3592	147
HotpotQA	512	Yes	0.520	0.400	3768	150
HotpotQA	1024	No	0.560	0.430	6164	150
HotpotQA	1024	Yes	0.540	0.389	7280	150
SQuAD v2	256	No	0.987	0.726	2255	150
SQuAD v2	256	Yes	0.967	0.726	1852	150
SQuAD v2	512	No	1.000	0.714	2350	150
SQuAD v2	512	Yes	1.000	0.722	2156	150
SQuAD v2	1024	No	1.000	0.733	2298	150
SQuAD v2	1024	Yes	1.000	0.733	2178	150

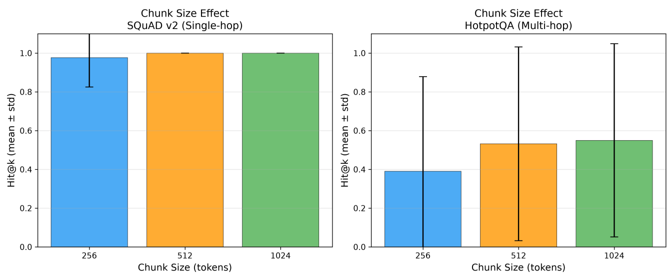

Fig. 1. Hit@k by chunk size (mean $\pm$ std). SQuAD v2 is near-ceiling at every chunk size; HotpotQA shows a clear upward trend from 256 to 1024 tokens.

- **Answer F1 (answer quality).** Token-level F1 between generated and gold answers, with standard SQuAD-style normalization.
- **Latency.** Wall-clock time per question from chunking through generation, in milliseconds.

E. Statistical Testing

For the reranking factor, we test the null hypothesis of no effect using a paired Wilcoxon signed-rank test, paired at the (seed, chunk size) level, significance threshold $\alpha = 0.05$.

IV. RESULTS

A. Main Effects

Three of the intended 1800 question-level runs did not complete (HotpotQA, chunk=512, no-rerank, $n = 147/150$) due to transient generation errors; see Limitations.

Chunk size. On HotpotQA, larger chunks substantially improve retrieval: Hit@k rises from 0.38–0.40 at 256 tokens to 0.54–0.56 at 1024 tokens—the largest effect of any factor in this study. On SQuAD v2, Hit@k is already at or near ceiling (0.97–1.00) at every chunk size.

Reranking. Table I shows small, inconsistent differences between reranked and non-reranked conditions on both datasets. Reranking consistently *increases* latency, roughly doubling it at the 256-token chunk size on HotpotQA (1186ms $\rightarrow$ 2096ms).

TABLE II
PAIRED WILCOXON TEST: RERANKING ON VS. OFF

Dataset	Rerank	No Rerank	Diff	p	Sig.
SQuAD v2	0.989	0.996	-0.007	0.500	No
HotpotQA	0.487	0.494	-0.008	0.656	No

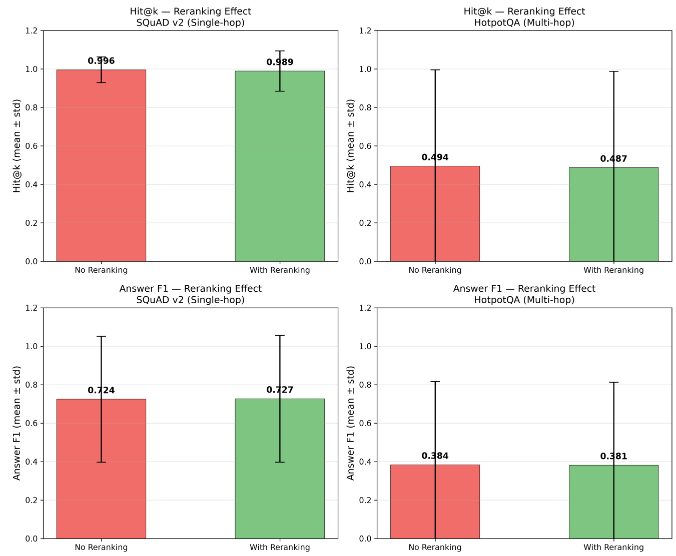

Fig. 2. Hit@k (top) and Answer F1 (bottom) with and without reranking. Differences are small and within error bars on both datasets.

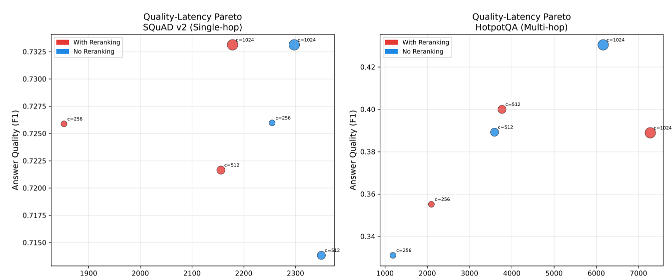

Fig. 3. Answer F1 vs. latency, colored by reranking condition, sized by chunk size. Reranking does not appear on the efficient frontier at any chunk size in this study.

B. Significance of the Reranking Effect

Reranking produces no statistically significant change in retrieval quality on either dataset. On SQuAD v2, this is unsurprising given the ceiling effect. On HotpotQA, where Hit@k is well below ceiling and reranking would plausibly have room to help, we still find no significant effect ($p = 0.656$), and the point estimate is directionally negative.

C. Quality–Latency Trade-off

Across both datasets, reranking's cost is consistent and its benefit is not: it adds 40–100%+ to per-question latency without a corresponding, statistically detectable gain. Chunk size, by contrast, is largely "free" in latency terms on SQuAD, and only moderately increases latency on HotpotQA.

V. Discussion

The central finding of this study is a negative result relative to our working hypothesis: we did not find reranking to help more on multi-hop questions than on single-hop questions—or, within the power of this study, to help significantly at all. This runs counter to the common assumption that cross-encoder reranking should matter more when relevant evidence is scattered across multiple passages.

Two explanations are worth separating. First, it is possible reranking genuinely does not help here—bi-encoder retrieval followed by simple top-k truncation may already surface adequate context. Second, it is possible the effect exists but is too small to detect at this sample size ($n \approx 150$ per condition).

What the data show more clearly is that chunk size, not reranking, is the dominant lever for multi-hop retrieval quality in this pipeline: moving from 256 to 1024 tokens nearly doubled effective retrieval performance on HotpotQA. This suggests a simple, actionable practitioner takeaway: before reaching for a reranker, check whether chunks are simply too small to contain the evidence a multi-hop question needs.

Practitioner decision rule. For single-hop, factoid-style retrieval, skip reranking—retrieval is already near-ceiling and reranking only adds latency. For multi-hop retrieval, prioritize increasing chunk size before adding reranking; neither dataset showed a statistically detectable benefit from reranking at any chunk size, so reranking should be treated as an optional, unproven latency cost rather than a default component, pending further study at larger sample sizes.

VI. Limitations

1) **Single embedding model.** The originally planned free-vs-paid embedding comparison (BAAI/bge-small-en-v1.5 vs. OpenAI text-embedding-3-small) was dropped from the final sweep because no billing was configured for the OpenAI account during the study window. This leaves the embedding-choice question empirically untested.
2) **Incomplete runs.** 3 of 1800 targeted observations did not complete due to transient errors ($n = 147$ instead of 150 in one cell).
3) **Single generator model.** All generation used Llama-3.1-8B-Instant via Groq's free tier. Retrieval-quality findings (Hit@k) are generator-independent; answer-quality (F1) findings may not generalize to other generators.
4) **Retrieval metric granularity.** Hit@k is an exact-substring match between the gold answer and retrieved text, a coarser proxy than gold-passage-identity matching or LLM-judged relevance.
5) **Sample size and statistical power.** With $\sim$150 observations per condition, this study can detect only moderate-to-large effects; small reranking effects (on the order of 1–2 percentage points) cannot be ruled out.

Acknowledgment

This project was completed as part of a research-track candidate assessment for Anvoxa Research.

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